

JEE MAIN REVISION MASTER: ALCOHOLS, PHENOLS AND ETHERS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 60 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Acidity of Phenols & Distinguishing Tests	30 PYQs	Phenols
Very High	Williamson Ether Synthesis & Cleavage by HI	30 PYQs	Ethers

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Acidity of Phenols & Distinguishing Tests	$\text{Acidity: Picric acid} > \text{p-Nitrophenol} > \text{Phenol} > \text{Water} > \text{Ethanol}$, $\text{Lucas: } 3^\circ \text{ (turbidity immediately)}, 2^\circ \text{ (5 mins)}, 1^\circ \text{ (no turbidity)}$
Williamson Ether Synthesis & Cleavage by HI	$\text{Williamson: } \text{R-O}^- + \text{Na}^+ + \text{R'-X (must be } 1^\circ) \rightarrow \text{R-O-R'} + \text{NaX}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Acidity of Phenols & Distinguishing Tests (Very High Yield — 30 PYQs)

Core Principle: Phenoxide resonance stabilization, substituent effects on acidity, Reimer-Tiemann and Kolbe-Schmitt reactions, and Lucas test.

■ **Caution / JEE Pitfall:** Electron withdrawing groups (-NO₂, -CN) INCREASE phenol acidity (especially at ortho and para positions), while electron donating groups (-CH₃) decrease acidity.

■ **Speed Trick:** Reimer-Tiemann reaction involves dichlorocarbene (:CCl₂) as an electrophilic intermediate to convert phenol to salicylaldehyde.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "Consider the following reactions. Total number of electrons in the π bonds and lone pair of electrons in the product (X) is :..." [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "The reactions which produce alcohol as the product are : A. $\text{CH}_3\text{CHO} + \text{O}_2 \xrightarrow{\text{MoO}_3, \Delta} \text{CH}_3\text{COOH}$ B. $2\text{CH}_3\text{CHO} \xrightarrow{\text{B}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ " [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "A mixed ether (P), when heated with excess of hot concentrated hydrogen iodide produces two different alkyl iodides which when treated wi..." [Ans: N/A]

3.2 Williamson Ether Synthesis & Cleavage by HI (Very High Yield — 30 PYQs)

Core Principle: Alkoxide SN₂ displacement on primary halides, and ether cleavage mechanism with concentrated HI.

■ **Caution / JEE Pitfall:** In Williamson synthesis, the alkyl halide **MUST** be primary (1°). If a tertiary (3°) halide is used, elimination dominates to give an **ALKENE** instead of an ether!

■ **Speed Trick:** With unsymmetrical ethers + HI: If one alkyl group is tertiary (3°), the iodide attaches to the 3° carbon (via SN₁). If both are 1° or 2°, iodide attacks the less hindered carbon (via SN₂).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Consider the following reaction. Statement I : In the above reaction, product formed will be a mixture of benzyl alcohol and iodobenzene..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Identify the colour of compound 'X' in the sequence of the reaction..." **[Ans: N/A]**
- **[2026 Morning Shift] Q#1 (MCQ):** "A hydroxy compound (X) with molar mass 122 g mol⁻¹ is acetylated with acetic anhydride, using a large excess of the reagent ensuring complete reaction..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Identify the colour of compound 'X' in the sequence of the reaction.

- (A) Violet
- (B) Green
- (C) Red
- (D) Colourless

Correct Answer: None

Solution Analysis: Compound X is phenolphthalein (an acid–base indicator). In basic solution, phenolphthalein first shows a pink colour. But when NaOH is in excess (very strong base), phenolphthalein changes to a form that is colourless. So, the colour of compound X in excess NaOH is colourless.

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#2)

The reactions which produce alcohol as the product are : A. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + \text{O}_2 \xrightarrow{\text{MoO}_3, \Delta} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ B. $2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + 3\text{O}_2 \xrightarrow{(\text{CH}_3\text{COO})_2\text{Mn}, \Delta} 2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{CH}_3\text{COOH}$ C. $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_3 \xrightarrow{\text{KMnO}_4} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{CH}_3\text{COOH}$ D. $2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + \text{O}_2 \xrightarrow{\text{Cu}, 523\text{K}, 100\text{atm}} 2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{H}_2\text{O}$ E. $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3 \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ Choose the correct answer from the options given below :

- (A) , D and E Only
- (B) and D Only
- (C) , C and E Only
- (D) and D Only

Correct Answer: None

Solution Analysis: Reaction given Alcohol (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + \text{O}_2 \xrightarrow{\text{MoO}_3, \Delta} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ (B) $2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + 3\text{O}_2 \xrightarrow{(\text{CH}_3\text{COO})_2\text{Mn}, \Delta} 2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{CH}_3\text{COOH}$ (D) $2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 + \text{O}_2 \xrightarrow{\text{Cu}, 523\text{K}, 100\text{atm}} 2\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{H}_2\text{O}$ And. $\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_3 \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$